

CUSTOMER SEGMENTATION & CHURN PATTERN ANALYTICS

in European Retail Banking

A Data Analytics Research Project

Prepared for Unified Mentor | Subject: The European Central Bank

Prepared by: Taha S. Travadi
GEC Bhavnagar / Computer Engineering

October 2026

Table of Contents

Table of Contents2

Abstract3

1. Introduction.....3

2. Problem Statement3

3. Project Objectives3

3.1 Primary Objectives3

3.2 Secondary Objectives4

4. Dataset Description.....4

5. Methodology4

5.1 Data Ingestion & Validation4

5.2 Data Cleaning & Feature Engineering.....5

5.3 Tools & Techniques5

6. Exploratory Data Analysis & Key Findings5

6.1 Overall Churn Rate5

6.2 Geographic Churn Patterns.....5

6.3 Age Group Analysis — Rate vs. Volume5

6.4 Tenure — A Non-Factor6

6.5 Gender-Based Differences.....6

6.6 Balance Segment — The High-Value Paradox.....6

7. High-Value Customer & Revenue Risk Analysis6

7.1 Geographic & Age Concentration of High-Value Churn7

8. Interactive Power BI Dashboard7

8.1 Page 1 — Overall Churn Summary7

8.2 Page 2 — Geography-wise Churn Visualization8

8.3 Page 3 — Age & Tenure Churn Comparison9

8.4 Page 4 — High-Value Customer Churn Explorer9

9. Key Insights & Business Implications10

10. Recommendations.....10

11. Conclusion10

12. Limitations & Future Scope.....10

Abstract

Customer churn is one of the largest hidden costs in retail banking, eroding lifetime value, inflating acquisition costs, and destabilising revenue. This project analyses a 10,000-customer dataset from a European retail bank operating across France, Germany, and Spain to uncover which customer segments are most at risk of churning and why. Using Python for exploratory data analysis and Power BI for an interactive, filterable dashboard, the project segments customers across five dimensions — geography, age, credit score, tenure, and account balance — and quantifies churn behaviour within each.

The analysis finds an overall churn rate of 20.37%, but this figure masks sharp disparities beneath it. Germany churns at roughly double the rate of France and Spain (32.44% vs ~16%), with the 46–60 age group across all three countries churning at the highest rate (51.12%). Counter-intuitively, high-balance customers churn more often than low- or zero-balance customers, and female customers churn at a substantially higher rate than male customers (25.07% vs 16.46%). Tenure, by contrast, shows almost no relationship with churn. Combining these findings, the project quantifies revenue at risk: churned high-value customers held approximately €185.6 million in balances — 29.21% of all high-value balances in the dataset — with German customers aged 30–45 alone accounting for 27% of all high-value churners.

These findings are translated into a set of targeted, segment-specific retention recommendations for the bank, delivered alongside a live, filterable Power BI dashboard that allows stakeholders to explore churn risk by geography, age, gender, and balance segment in real time.

1. Introduction

Retail banks accumulate vast amounts of customer-level data, yet many struggle to translate that data into actionable retention strategy. Churn — the loss of an existing customer — is typically tracked as a single aggregate rate, which tells a bank that it is losing customers but not which customers, why, or what it is costing them. This project was undertaken to close that gap for a simulated European retail bank operating in France, Germany, and Spain, using a dataset of 10,000 customers containing demographic, financial, and behavioural attributes alongside a binary churn indicator.

The guiding philosophy of this project is that churn is not a single phenomenon but a collection of distinct risk patterns that only become visible once the customer base is properly segmented. A bank that treats a 25-year-old low-balance customer in Spain the same as a 55-year-old high-balance customer in Germany, for retention purposes, is almost certainly misallocating its retention budget. This project builds the segmentation framework needed to tell those two customers apart.

2. Problem Statement

Despite having rich customer-level data, the bank faces three concrete analytical gaps:

- Identifying which customer segments carry disproportionately high churn risk.
- Understanding how churn differs across geography, age, gender, and financial profile.
- Quantifying the financial exposure created by churn among high-value customers.

Without structured, segmentation-driven analytics, retention strategy remains generic, reactive, and inefficient — resources are spread evenly across a customer base that, as this project demonstrates, carries highly uneven risk.

3. Project Objectives

3.1 Primary Objectives

- Measure the overall churn rate of the customer base.
- Identify how churn is distributed across defined customer segments.

- Compare churn behaviour across the bank's three operating geographies.

3.2 Secondary Objectives

- Understand churn specifically among high-value (high-balance) customers.
- Evaluate the relationship between engagement, tenure, and churn.
- Produce a live, interactive tool that supports ongoing strategic and marketing decisions.

4. Dataset Description

The dataset consists of 10,000 customer records with 14 original fields. Each row represents one bank customer, with a binary target variable, *Exited*, indicating whether that customer churned (1) or was retained (0).

Column	Description
CustomerId	Unique customer identifier
Surname	Customer surname (dropped during cleaning — not analytically useful)
CreditScore	Customer creditworthiness (range observed: 350–850)
Geography	Customer's country: France, Spain, or Germany
Gender	Male / Female
Age	Customer age in years
Tenure	Years the customer has banked with the institution
Balance	Customer's account balance
NumOfProducts	Number of bank products held
HasCrCard	Credit card ownership (binary)
IsActiveMember	Activity indicator (binary)
EstimatedSalary	Customer's estimated annual salary
Exited	Churn indicator — target variable (1 = churned, 0 = retained)

Table 1: Dataset field descriptions

A fourteenth field, *Year*, was present in the raw file but found to be a constant value (2025) across all 10,000 rows during validation, and was therefore dropped as analytically redundant.

5. Methodology

5.1 Data Ingestion & Validation

The dataset was loaded in Python (pandas) and validated for structural integrity before any analysis began. This included confirming there were no missing values or duplicate rows, verifying that *CustomerId* values were all unique, confirming that *Geography* and *Gender* contained only their expected categories, and confirming that binary fields (*HasCrCard*, *IsActiveMember*, *Exited*) contained only 0/1 values. *CreditScore* was confirmed to fall within a realistic 350–850 range. This validation step established that the dataset required no corrective cleaning of invalid values — only the removal of non-analytical and constant fields.

5.2 Data Cleaning & Feature Engineering

The Surname and Year fields were dropped. Four new segmentation fields were then engineered from the existing continuous variables, directly supporting the segmentation design required by the project brief:

- AgeGroup: <30, 30–45, 46–60, 60+
- CreditBand: Low, Medium, High (banded using standard FICO-style thresholds)
- TenureGroup: New, Mid-term, Long-term
- BalanceSegment: Zero-Balance, Low-Balance, High-Balance

Each engineered segment was checked for population balance to confirm no segment was empty or negligibly small, ensuring every downstream comparison was statistically meaningful.

5.3 Tools & Techniques

Exploratory data analysis, segmentation logic, and statistical summarisation were performed in Python using pandas, matplotlib, and seaborn. The interactive dashboard was built in Power BI Desktop using DAX measures for all KPIs, enabling dynamic recalculation as a user applies filters. This dual-tool approach was chosen deliberately: Python for rigorous, reproducible analysis, and Power BI for a live, stakeholder-facing exploration tool — reflecting the two skill sets most relevant to a modern banking analytics function.

6. Exploratory Data Analysis & Key Findings

6.1 Overall Churn Rate

Of the 10,000 customers in the dataset, 2,037 churned, giving an overall churn rate of 20.37%. This is the baseline figure against which every segment below is compared — a segment materially above 20.37% represents elevated risk; a segment materially below represents relative stability.

6.2 Geographic Churn Patterns

Churn is heavily concentrated in Germany. While France (16.15%) and Spain (16.67%) churn at similar, below-average rates, Germany churns at 32.44% — exactly double its peers, despite holding a comparable share of the overall customer base. Geography is, on this evidence, the single strongest standalone predictor of churn risk in the dataset.

Geography	Churn Rate
France	16.15%
Germany	32.44%
Spain	16.67%
Overall	20.37%

Table 2: Churn rate by geography

6.3 Age Group Analysis — Rate vs. Volume

Churn rises sharply with age up to the 46–60 band before falling again, rather than increasing linearly. This is an important distinction with two separate practical implications: which segment is riskiest per customer (rate), and which segment contributes the most churned customers overall (volume).

Age Group	Churn Rate	Customer Count
<30	7.52%	1,968
30–45	15.74%	5,921
46–60	51.12%	1,647
60+	24.78%	464

Table 3: Churn rate and customer volume by age group

The 46–60 age group carries by far the highest individual risk (51.12% — more than two and a half times the overall rate), making it the segment with the greatest need for proactive retention outreach per customer. The 30–45 group, however, is nearly four times larger in absolute size, meaning it still contributes the largest total number of churned customers to the bank's overall losses. A retention strategy should treat these as two distinct priorities: 46–60 for highest-urgency individual intervention, 30–45 for highest-volume campaign reach.

6.4 Tenure — A Non-Factor

Contrary to the common assumption that customer loyalty builds protection over time, tenure showed almost no relationship with churn: New customers (0–2 years) churned at 21.15%, Mid-term customers at 20.76%, and Long-term customers at 19.67% — a spread of only 1.48 percentage points across the entire relationship lifecycle. This suggests that, for this bank, retention risk is not primarily a function of relationship length, and that tenure-based loyalty programmes alone are unlikely to meaningfully reduce churn without being paired with the other risk factors identified in this analysis.

6.5 Gender-Based Differences

Female customers churn at a substantially higher rate than male customers — 25.07% versus 16.46%, a gap of 8.6 percentage points. This gap is not uniform across geography: it widens considerably in Germany (female 37.55% vs male 27.81%) and is comparatively narrower in Spain (21.21% vs 13.11%). Germany therefore represents the point of maximum combined risk — where the geography effect and the gender effect compound.

Gender	France	Germany	Spain	Overall
Female	20.34%	37.55%	21.21%	25.07%
Male	12.71%	27.81%	13.11%	16.46%

Table 4: Churn rate by gender and geography

6.6 Balance Segment — The High-Value Paradox

The most counter-intuitive finding of this project is that high-balance customers churn more frequently than low- or zero-balance customers, not less. Conventional banking intuition holds that a customer's larger balance should signal stronger loyalty; the data shows the opposite. This single finding reframes the entire retention priority: the bank's most valuable customers, in balance terms, are also disproportionately its flight-risk customers, and standard retention effort directed at low-engagement, low-balance customers may be systematically misdirected.

7. High-Value Customer & Revenue Risk Analysis

Building on the balance paradox above, high-value customers (defined here as customers in the High-Balance segment) were isolated for a dedicated risk analysis, since this is the segment with the greatest direct financial exposure per lost customer.

High-Value Churn Rate 25.23%	Revenue at Risk €185.59M	% of High-Value Balance at Risk 29.21%	Total High-Value Balance €635.34M
--	------------------------------------	---	---

High-value customers churn at 25.23% — nearly 5 points above the overall rate of 20.37% — confirming that this is not a marginal effect but a structurally elevated risk group. In balance terms, customers who churned while holding a high-balance account represented €185.59 million, or 29.21% of all balances held by high-value customers in the dataset. Put plainly: almost three out of every ten euros held by the bank's wealthiest customer segment is sitting with a customer who ultimately left.

7.1 Geographic & Age Concentration of High-Value Churn

Breaking high-value churners down by geography and age group reveals where this revenue risk is concentrated. Germany alone accounts for 58.30% of all high-value churners (706 of 1,211), despite representing roughly a third of the overall customer base — a striking over-representation. Within that, the single largest concentration cell in the entire matrix is German customers aged 30–45, who alone represent 27.00% of all high-value churners.

Geography / Age Group	<30	30–45	46–60	60+	Total
France	1.90%	13.46%	11.15%	1.82%	28.32%
Germany	4.38%	27.00%	23.95%	2.97%	58.30%
Spain	2.06%	5.20%	5.28%	0.83%	13.38%
Total	8.34%	45.66%	40.38%	5.62%	100.00%

Table 5: Share of all high-value churners, by geography and age group

More than one in four high-value customers who churned was a German customer between the ages of 30 and 45 — the single highest-priority segment for high-value retention effort identified anywhere in this analysis.

8. Interactive Power BI Dashboard

To make these findings explorable rather than static, a four-page interactive Power BI dashboard was built, mirroring the analytical structure of this report. Every page shares a synchronised set of filters (Geography, Gender, Age Group, Balance Segment), so a user can narrow the entire dashboard — KPIs and charts across all four pages — to any customer subset of interest.

8.1 Page 1 — Overall Churn Summary

Presents headline KPIs (Total Customers, Churn Rate, High-Value Churn Rate, Revenue at Risk) alongside the retained-vs-churned split and churn rate by gender.

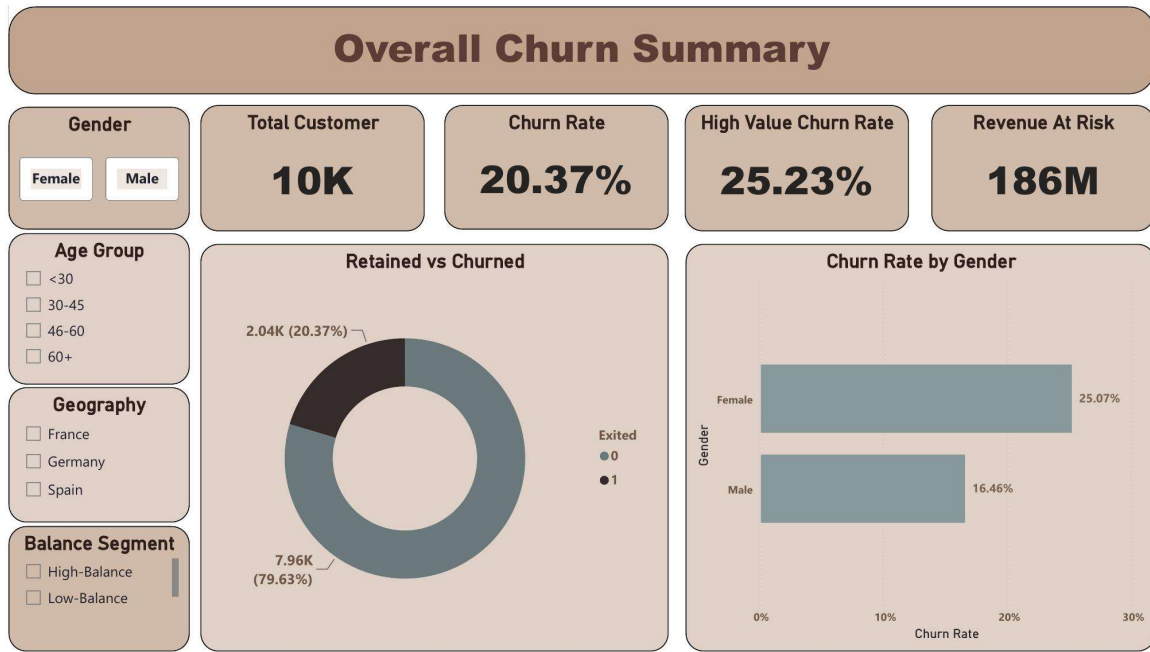


Figure 1: Overall Churn Summary page

8.2 Page 2 — Geography-wise Churn Visualization

Presents a colour-scaled heatmap and matrix of churn rate by geography and age group, plus a secondary matrix crossing geography and gender — surfacing the Germany/46-60 concentration and the gender-geography interaction described in Section 6.

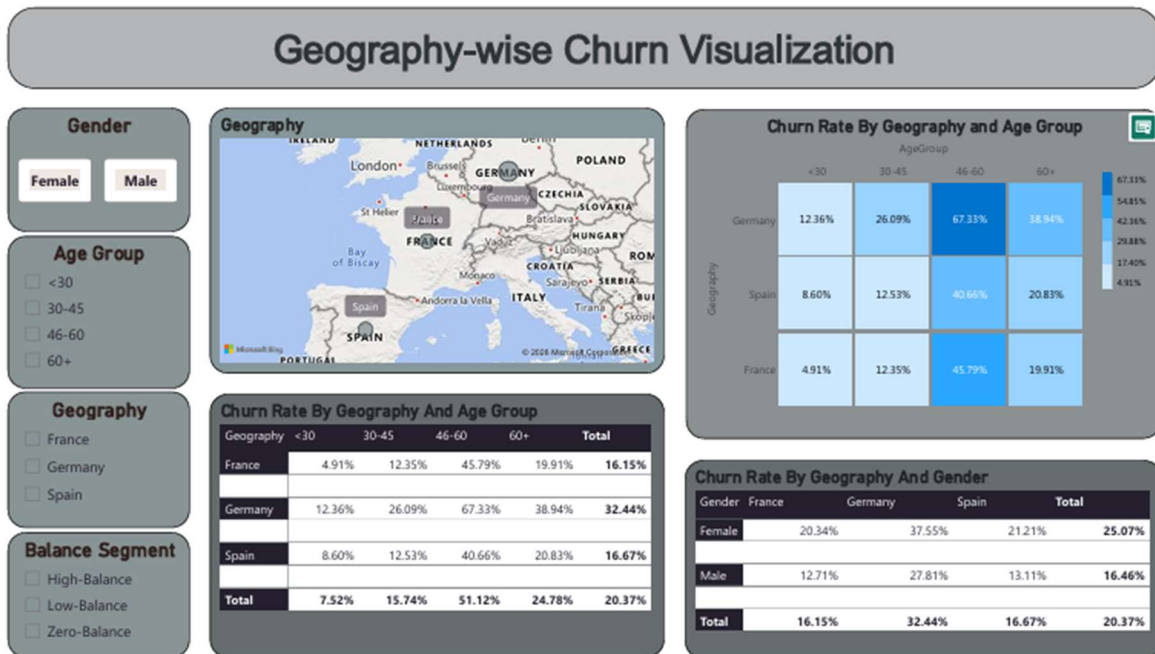


Figure 2: Geography-wise Churn Visualization page

8.3 Page 3 — Age & Tenure Churn Comparison

Presents churn rate by age group and tenure group side by side, alongside a combination chart overlaying customer count (bars) and churn rate (line) by age group — visually demonstrating the rate-versus-volume distinction central to Section 6.3.

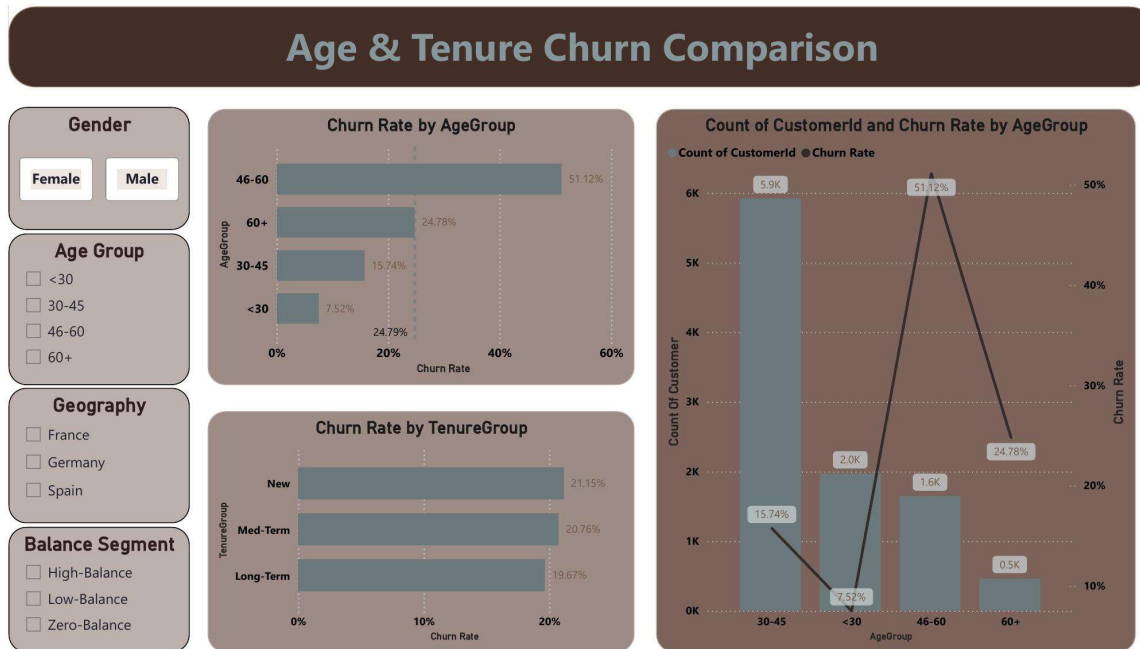


Figure 3: Age & Tenure Churn Comparison page

8.4 Page 4 — High-Value Customer Churn Explorer

Presents the high-value KPI set (High-Value Churn Rate, Revenue at Risk, % Revenue at Risk, Total High-Value Balance), an age-versus-balance scatter plot coloured by churn status, and the full geography-by-age-group high-value churn matrix from Table 5.

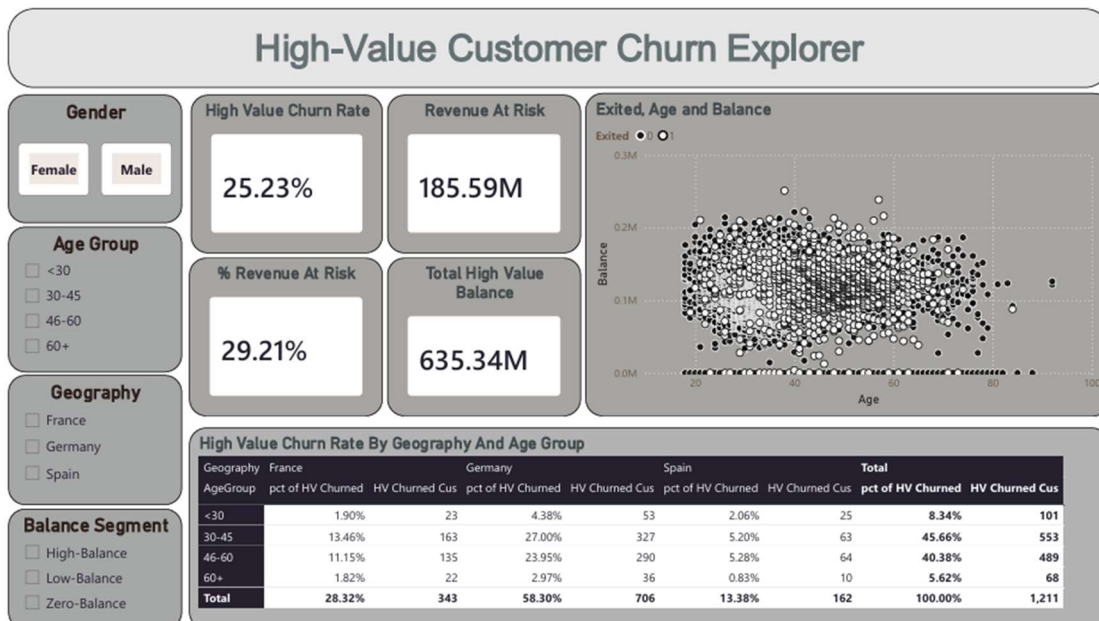


Figure 4: High-Value Customer Churn Explorer page

9. Key Insights & Business Implications

- Germany is the bank's single largest churn risk market — churning at double the rate of France and Spain, and contributing well over half of all high-value churners. Any retention budget reallocation should weight Germany disproportionately.
- The 46–60 age group is the highest individual risk segment (51.12% churn rate), suggesting a life-stage or product-fit issue specific to this cohort — possibly retirement planning needs, competitor offers aimed at this demographic, or product dissatisfaction that accumulates with account age.
- High-balance customers are a structurally elevated risk group, not a protected one. This likely reflects that wealthier customers have greater means and motivation to shop around for better rates or service elsewhere — the bank cannot assume balance size equals loyalty.
- Tenure provides little to no protective effect. Loyalty-tenure-based retention programmes (e.g. "thank you for 5 years" campaigns) are unlikely, on their own, to move the churn needle.
- Female customers churn notably more than male customers, especially in Germany — a gap worth investigating qualitatively (service experience, product fit, marketing relevance) rather than assuming it is explained by the other variables already measured.
- German customers aged 30–45 are the single highest-priority segment for high-value retention spend, representing more revenue-at-risk concentration than any other geography-age combination.

10. Recommendations

- Launch a Germany-specific retention programme, prioritising the 30–45 and 46–60 age bands, with particular attention to high-balance accounts — this single intersection represents the bank's largest concentrated revenue exposure.
- Build an early-warning model flagging high-balance customers aged 46–60 for proactive relationship-manager outreach before churn signals (e.g. inactivity) appear, given this group's outsized individual risk.
- Investigate the gender churn gap qualitatively — through customer surveys or exit interviews — particularly in Germany, to determine whether it reflects a product, pricing, or service experience gap rather than treating it as a fixed demographic pattern.
- Reassess the strategic value of tenure-based loyalty incentives, redirecting that budget toward the segment-specific interventions above, given tenure's demonstrated lack of correlation with retention.
- Deploy the accompanying Power BI dashboard to relationship managers and regional leadership as a live monitoring tool, allowing teams to track these segments on an ongoing basis rather than relying on static quarterly churn reports.

11. Conclusion

This project set out to move the bank's understanding of churn beyond a single aggregate rate and toward a segmentation-driven view of where risk actually concentrates. The analysis shows clearly that churn in this dataset is not evenly distributed: it is sharply concentrated in Germany, in the 46–60 age band, among high-balance customers, and — more than many banks might assume — among their most financially valuable customers rather than their least. With €185.6 million in high-value balances tied to churned accounts, the financial stakes of acting on these findings are substantial. The accompanying interactive dashboard ensures these insights remain usable beyond this report — as a live tool for ongoing, segment-aware retention decision-making.

12. Limitations & Future Scope

- The dataset is a single snapshot in time (Year = 2025); it does not capture churn trends over multiple years, which would allow seasonality or trend analysis.
- No direct cause of churn (e.g. complaint records, competitor switching reason) is captured — the analysis identifies who churns and in what combinations, not the underlying motivation, which would require qualitative follow-up.

- High-value customer status was defined using account balance alone for consistency across both the Python and Power BI analyses; incorporating salary or product holdings into a composite value score is a natural next iteration.
- A predictive model (e.g. logistic regression or gradient boosting) trained on these segments could extend this project from descriptive analytics to a working churn-prediction tool, enabling individual customer risk scoring rather than segment-level risk only.